Closed-Form Orbits of the Maps $ax + 1$: A Characterisation of Fermat Bases

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Abstract

For odd a consider the accelerated map $x \mapsto \text{odd_part}(ax+1)$ on odd integers, and the associated "block" parametrisation $x = 2^{r+1}m - 1$. We characterise exactly those bases a for which the iterates admit the closed polynomial form $x_i = (a^i m - 1)/(a-1)$: this happens *if and only if* $a - 1$ is a power of 2, i.e. $a = 2^e + 1$ — a Fermat-type base $(3, 5, 9, 17, 33, \dots)$. Collatz is the case $a = 3$. Both directions follow from a single one-step law, valid for every odd a. **The whole characterisation — the one-step law, the biconditional, and the orbit formula — is formalised in Lean 4 and machine-checked with kernel-only axioms. This is a theorem about iteration of affine maps and is independent of the Collatz conjecture.**

MSC: 37P05 (Arithmetic dynamics), 11B37 (Recurrences), 68V20 (Formalisation of mathematics).

1 Setting

Fix an odd integer $a \geq 3$. On the odd integers define

$$\text{step}_a(x) = \frac{ax+1}{2^{v_2(ax+1)}},$$

where v_2 is the 2-adic valuation. Every odd x can be written uniquely as

$$x = 2^{r+1}m - 1, \qquad m \text{ odd}, \ r \geq 0, \tag{1}$$

so that $r = v_2(x+1) - 1$ records how many leading valuation-one steps the orbit begins with.

The question we answer is: *for which a do the iterates of* step_a *starting from* (1) *follow a single closed-form polynomial law?*

2 The characterisation

Theorem 2.1 (One-step law, and the characterisation)**.** *Let $a \geq 3$ be odd and write $a - 1 = 2^e q$ with q odd. For every odd m and every $r \geq e$, put $x = 2^{r+1}m - 1$. Then*

$$v_2(ax+1) = e \qquad \textit{and} \qquad \text{step}_a(x) = a\,2^{\,r+1-e}m - q. \tag{2}$$

Hence the closed-form law $\text{step}_a(x) = a\,2^{\,r+1-e}m - 1$ *holds* if and only if $q = 1$, *i.e. iff $a - 1$ is a power of 2, i.e. iff $a = 2^e + 1$. In that case iteration gives*

$$x_i = a^i\,2^{\,r+1-ie}m - 1 \qquad \textit{for every } i \textit{ with } ie \leq r. \tag{3}$$

Proof. Compute:

$$ax + 1 = a(2^{r+1}m - 1) + 1 = a\,2^{r+1}m - (a-1) = 2^e\big(a\,2^{r+1-e}m - q\big).$$

As $r \geq e$, the bracket is (even)−(odd), hence odd; so $v_2(ax+1) = e$ *exactly* and $\mathrm{step}_a(x)$ is the bracket, which is (2). Its value equals $a2^{r+1-e}m - 1$ precisely when $q = 1$ — both directions at once.

For (3), let $q = 1$. The image $a2^{r+1-e}m - 1$ has again the shape (1), with odd parameter $m' = am$ and exponent $r' + 1 = r + 1 - e$; so the family is preserved and induction on i gives (3) while $ie \leq r$. □

The one-step law (2) holds for *every* odd a; the characterisation is simply the observation that its right-hand side falls back into the family exactly when the odd part q of $a - 1$ is trivial.

a	$a - 1 = 2^e q$	q	closed form?
3	$2^1 \cdot 1$	1	yes (Collatz)
5	$2^2 \cdot 1$	1	yes
7	$2^1 \cdot 3$	3	no
9	$2^3 \cdot 1$	1	yes
11	$2^1 \cdot 5$	5	no
13	$2^2 \cdot 3$	3	no
15	$2^1 \cdot 7$	7	no
17	$2^4 \cdot 1$	1	yes
33	$2^5 \cdot 1$	1	yes

Table 1: The closed form holds exactly when the odd part q of $a - 1$ is 1. The one-step law and the closed form were tested independently by running the map — the "no" column is a genuine test, not a restatement of the hypothesis — for all odd $a \leq 999$.

Remark 2.2 (Terminology)**.** The bases $a = 2^e + 1$ are $3, 5, 9, 17, 33, 65, 129, 257, 513, \ldots$; we call them *Fermat-type* because the Fermat numbers $2^{2^k} + 1$ are the sub-family $e = 2^k$. The two sets are *not* equal: $9, 33, 65, 129, 513$ are of the form $2^e + 1$ but are not Fermat numbers.

3 Formalisation

The orbit formula (3) is formalised in Lean 4 as `orbit_formula_fermat`:

$$\texttt{iter_a}\ a\ i\ (2^{r+1}m - 1) = a^i\,2^{r+1-ie}m - 1 \qquad (ie \leq r),$$

for $a = 2^e + 1$, $e \geq 1$, m odd. It builds with

```
lake build CollatzProof.KodamaFermat
```

and `#print axioms` reports exactly the three ambient kernel axioms (`propext`, `Classical.choice`, `Quot.sound`) — no `sorry` and no custom axioms. The audit lives in `CollatzProof/AxiomAudit.lean` so that it re-runs on every build.

Since 2026-07-17 the *whole* characterisation is in the kernel, not just one half:

- `step_a_general`: the one-step law (2) for *every* odd base a — write $a - 1 = 2^e q$ with q odd, then $v_2(an+1) = e$ and $\mathrm{step}_a(2^{s+e}m - 1) = a2^s m - q$;
- `closed_form_iff_fermat`: the biconditional $\mathrm{step}_a(2^{s+e}m - 1) = a2^s m - 1 \iff q = 1$;
- `fermat_iff_pow_two`: $a = 2^e + 1 \iff q = 1$.

All report exactly the three ambient kernel axioms. The audit `CollatzProof/AxiomAudit.lean` re-runs `#print axioms` on every one of them at each build, so the claim cannot silently rot.

4 Consequences and scope

Corollary 4.1. *The Collatz map ($a = 3$) is the smallest Fermat-type base; the next are $5, 9, 17, 33, \ldots$ (all such bases are odd, $2^e + 1$ being odd for $e \geq 1$). The heuristic drift of step_a is $\lambda_a = \log a - \mathbb{E}[v_2] \log 2 = \log a - 2 \log 2$, using $\mathbb{E}[v_2] = 2$ from the companion note [1], negative iff $a < 4$; thus $a = 3$ is the unique odd base on the contracting side and every odd $a \geq 5$ is super-critical.*

Remark 4.2 (What this does *not* say)**.** Theorem 2.1 is a statement about the algebraic shape of the iterates. It does not decide whether any particular orbit converges or diverges: the drift λ_a is an average and, as the divergence example $\lfloor \sqrt{2}\, n + 4 \rfloor$ of Althöfer–Zipproth shows, the sign of an average drift does not settle the fate of a deterministic orbit. In particular this paper says nothing about the Collatz conjecture, nor about the open question of whether $9x + 1$ diverges from $x = 1$.

Reproducibility

The base table is regenerated by `tools/verify_for_papers.py`; the Lean development builds with `lake build`.

Acknowledgement

AI-assisted; every claim was verified by exact computation or Lean's kernel before inclusion.

References

[1] J. Cruz Cabrera, *An exact equidistribution law for the 2-adic valuations of $3^r m - 1$*, companion note, 2026.

[2] J. C. Lagarias, *The $3x + 1$ problem and its generalizations*, Amer. Math. Monthly **92** (1985), 3–23.

[3] The mathlib Community, *The Lean mathematical library*, CPP 2020.

[4] I. Althöfer, T. Zipproth, *Divergence in a Collatz-type recursion*, preprint, August 2025.